cluster. Typically, these will run in their own Kubernetes namespace. In many cases, enabling TLS is then simply a case of adding some annotations to the deployment YAML files. The service mesh will then automatically inject the proxy sidecar container when your pods are started and configure them with TLS certificates.

In this section, you'll install the Linkerd service mesh and enable TLS between the Natter API, its database, and the link-preview service, so that all communicates are secured within the network. Linkerd has fewer features than Istio, but is much simpler to deploy and configure, which is why I've chosen it for the examples in this book. From a security perspective, the relative simplicity of Linkerd reduces the opportunity for vulnerabilities to be introduced into your cluster.

> DEFINITION The ***control plane*** of a service mesh is the set of components responsible for configuring, managing, and monitoring the proxies. The proxies themselves and the services they protect are known as the ***data plane***.

INSTALLING LINKERD

To install Linkerd you first need to install the `linkerd` command-line interface (CLI), which will be used to configure and control the service mesh. If you have Homebrew installed on a Mac or Linux box, then you can simply run the following command:

```
brew install linkerd
```